AI GATEWAY · PROMPT COMPRESSION & CONTEXT MANAGEMENT

Can we shorten the prompt without making the router worse?

The gateway asks an AI — “the judge” — to pick which model should answer each question. Our module shortens the text first. This is what that costs and what it saves.

`1,000 questions per test` `~12,700 judge calls` `0 errors` `judge: gpt-5.1-nano` `2026-09-11`

Two methods ship. One is dangerous. One is useless.

82%
TEXT REUSED FROM CACHE

−48%
TEXT IN LONG DOCUMENTS

−0.7
POINTS OF QUALITY, THAT COSTS

−40.8
POINTS IF WE SUMMARIZE CHATS

Reordering for cache is free and should go on everywhere. Shortening long documents costs well under a point. **Summarizing chat history destroys answers — 449 out of 996 — and must not ship.** An earlier version of this study said the opposite; it was testing on conversations where the history did not matter.

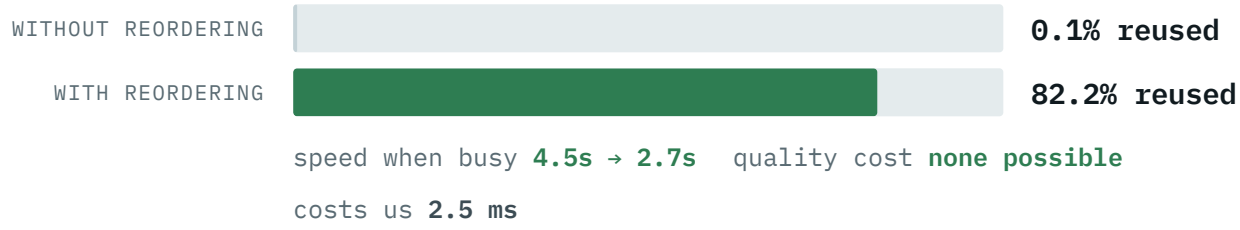
THE FOUR METHODS

What each one does, and whether to use it

1. Reorder the prompt so the cache can reuse it

TURN ON BY DEFAULT

Puts the parts that never change at the front and the changing part at the end. The AI provider only reuses text from the beginning of a prompt, so this is the difference between reusing almost everything and reusing nothing. It changes no words at all — only the order.



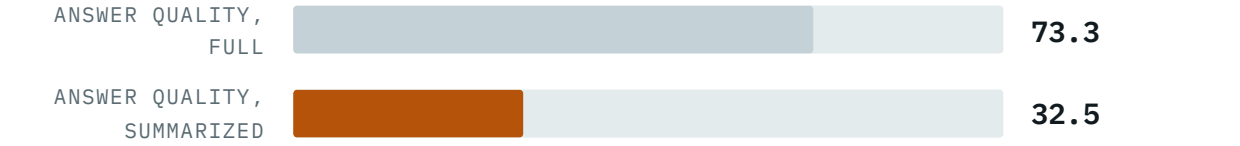
speed when busy **4.5s → 2.7s** quality cost **none possible**

costs us **2.5 ms**

2. Summarize old messages in long chats

DO NOT SHIP

Replaces old back-and-forth with a short recap, keeping the user's question and the last four exchanges word for word. It saves the most text of any method — and on conversations where the latest question *depends* on what was said earlier, it destroyed **449 of 996 answers**.

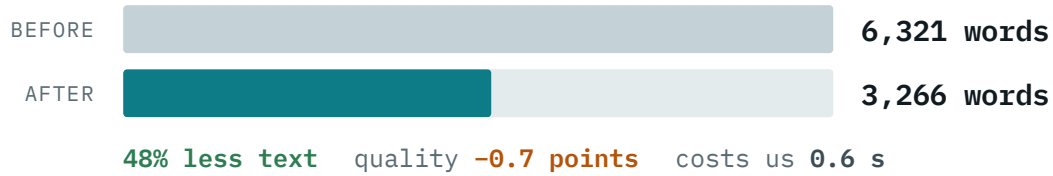


68% less text quality **-40.8 points** **449/996** answers destroyed

3. Shorten long documents

TURN ON ABOVE 5,000 WORDS

Uses a small model to drop low-value words from bulky pasted-in text. Only runs above 5,000 words — below that it makes requests slower rather than faster, so it declines. This is the one method that costs something.



48% less text quality **−0.7 points** costs us **0.6 s**

4. Reword the question more clearly

LEAVE OFF

Tries to rewrite the user's question, then compares it to the original and throws it away if the meaning drifted. It threw away its own work 86% of the time — the safety check working correctly on questions that have nothing to clarify.



nothing saved discarded its own work **86%** of the time wastes **1.3 s**

BEFORE READING ANY NUMBER

The judge disagrees with itself

We sent the **exact same 1,000 questions** through the judge twice, changing nothing at all. It sent **15% of them to a different model** the second time, and its overall score moved by 0.2 points.

> **So anything smaller than about 1.5 points is not a result.**
>
> Every difference on this page should be read against that. A method that scores 1 point worse has not been shown to be worse — the judge moves that much on its own. This is why we say a cost is *bounded* rather than *proven*.

DETAIL · SHORTENING LONG DOCUMENTS

How hard you shorten, and what it costs

1,000 questions, each wrapped in about 6,000 words of context. Every row reads the same questions; only the amount of shortening changes.

SETTING	TEXT THE JUDGE READS	QUALITY	CHANGE
No shortening REFERENCE	6,321	63.4	—
Same questions, run twice NOISE	6,321	63.3	−0.2
Our setting SHIPS	**3,266**	**62.7**	**−0.7**
Gentle	4,875	62.1	−1.3
Medium	3,154	61.7	−1.8
Aggressive	1,797	62.1	−1.4

Shortening more does not cost more in any clean way — aggressive scores better than medium. All of these sit within the judge's own wobble, which is the point: **we cut half the text and the quality difference is too small to pin down.**

DETAIL · REORDERING FOR CACHE

Free money, and it gets better under load

SITUATION	ORDER	REUSED FROM CACHE	SPEED
One at a time	**Ours**	**82.2%**	**2.76 s**
One at a time	Unordered	0.1%	2.83 s
8 at once	**Ours**	**79.9%**	**2.68 s**
8 at once	Unordered	7.6%	4.49 s

Sent one at a time the two orderings are nearly the same speed. Send eight at once — which is what a busy gateway looks like — and ours is **40% faster**. The benefit shows up exactly when you need it.

DETAIL · WHY SUMMARIZING FAILS

It does not just lose answers. It invents wrong ones.

996 real conversations where the last question **cannot be answered without the earlier ones** — “Where did she live?” needs to know who “she” is. Each was answered twice by the same model, once with the full conversation and once after summarizing.

QUESTION	CORRECT ANSWER	WITH FULL CHAT	AFTER SUMMARIZING
Did he want to take him for himself?	Yes	Yes	No
Do people prefer to buy things in Hong Kong?	no	No.	yes
What happened when Champion stepped off the bus?	He vomited.	He vomited.	He was beaten.
And LED?	30,000	30,000 hours	25,000 hours

61% of the damage is the model admitting it cannot answer. The other 39% is it answering confidently and wrongly.

A gateway that says “I don’t know” is annoying. A gateway that flips yes to no, or says someone was beaten when they vomited, is the kind of failure nobody catches until a customer does.

CORRECTIONS

Three things we said earlier that were wrong

Each was caught by testing something properly rather than by thinking harder about it. The third is the one that mattered.

1 **We recommended turning on chat summarizing. That was badly wrong.**

We reported **+0.9 points** and called it free. Our test conversations were built from unrelated questions, so summarizing them deleted nothing that mattered. On real conversations where the last question depends on the earlier ones, it costs **−40.8 points** and destroys 449 answers out of 996. The flaw was written down as a caveat and left unfixed for two rounds — a caveat nobody acts on is not much better than no caveat.

2 **We said document shortening was free. It is not.**

At 100 questions it measured exactly zero cost. At 1,000 it costs about 0.7 points, and **all five settings came out negative**. At the smaller size the results bounced between +2 and −1, which is what randomness looks like. Five out of five pointing the same way is not.

3 **We overstated how much shortening changed the judge's choices.**

We said it changed 16–18% of picks against a 2–7% baseline. But that baseline came from short questions and was applied to long ones. On long questions the judge changes **15% of its own picks** unprompted. Our real effect was 1–7%, not 10–16%.

LIMITS

What these numbers do not prove

! **The long documents and long chats are made up**

The benchmark contains neither, so we built them from unrelated questions. That is the *easy* case — there is little real meaning to destroy. Real retrieved documents are closely related to the question and much harder to shorten safely. Treat −75% and −48% as best cases.

! **For document shortening, we measured which model gets picked — not how good the final answer was**

The routing benchmark already knows how each model scores, so no model was asked to actually answer. Whether a *shortened document* produces a worse answer is still untested. That is the same gap that hid the summarizer's failure for two rounds, so it should not be left open.

! **Only the summarizer has been tested on realistic material**

The long documents are still built from unrelated text, which is the easy case. Real retrieved passages are closely related to the question. Given what happened when we tested the summarizer properly, the −48% figure deserves the same scrutiny before anyone relies on it.

RECOMMENDATION

What to do

1 **Turn on prompt reordering everywhere**

Free, cannot damage anything, and it both cuts cost and speeds things up under load. Nothing else in this study is this clearly worth it.

2 **Do not turn on chat summarizing**

It saves the most text of anything we built and it is the one thing here that is actively unsafe. Four fixes worth trying first: never summarize a turn holding a pasted document, keep far more recent turns word-for-word, summarize with the latest question in view, and skip summarizing when the latest question refers back to something.

3 **Keep document shortening switched off below 5,000 words**

Below that it slows requests down instead of speeding them up. Above it, half the text for well under a point.

4 **Test on real documents before quoting these savings outside the team**

This is the one thing that could change the answer. Real retrieved text is related to the question, and that is the case we have not measured.